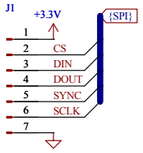
[8] Connectors

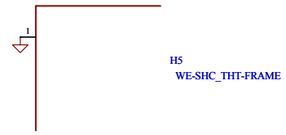
DESIGN NOTE:
J1 host header (JST-XH, 7 pins):
1 +3.3V 2 CS 3 DIN 4 DOUT/RDY
5 SYNC 6 SCLK 7 GND

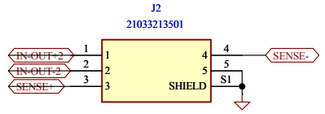
DESIGN NOTE:
J2/J4 load cell (5 pins):
1 OUT, 2 OUT (signal pair), 3 EXC+ (SENSE+),
4 EXC- (SENSE-/BPDSW), 5 GND/shield.
Both channels read V(pin2) - V(pin1):
fix the sign in firmware if needed.

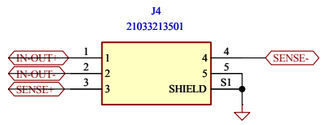
LAYOUT NOTE:
H5 shielding frame covers the AD7190
front-end; connector shells tied to GND.

Pinout and notes









Host and load-cell connectors





Power flags: +3.3V and GND come from the host connector J1,

Comments: **References:** JST XH B7B-XH-AM (J1) J2/J4 21033213501, H5 WE-SHC	Company: **OPEN THRUST LAB**		Variant:	Git Hash: 349b121
	Board Name: **AD7190**		Project Name: **OPEN_WEIGHT**	
Sheet Title: Connectors	File Name: Connectors.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-09-29	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Connectors/		Reviewer:	Size: **A3**	Sheet: **8** of **10**